

EPS 522/ENVS 423L Problem Set #7

Katie Markovich

Fall 2025

Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, November 13 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of ground-water processes.

1.1 Hydraulic Head

Given the following measurements at individual piezometers, calculate the hydraulic head, pressure head, elevation head, and the pressure. Assume the weight density of water (γ_w) is 9810 N/m^3 . (20 pts)

Piezometer	A	B	C	D	E
Surface elevation (m)	450	100	320	65	210
Piezometer depth (m)	150	100	80	120	20
Depth to water surface (m)	27	65	55	40	10
Hydraulic head (m)					
Elevation head (m)					
Pressure head (m)					
Pressure (N/m^2)					

1.2 Aquifer Storage

1.2.1 Unconfined

An aquifer has a specific yield of 0.19. During a drought period, the following average declines in the water table were noted:

Area	Size (km ²)	Decline (m)
A	15	2.34
B	7.5	1.22
C	18.3	0.76
D	22.5	3.44
E	9.44	1.89
F	22.7	0.35

What was the total volume of water represented by declines in the water table? (10 pts)

1.2.2 Confined

A confined aquifer has a specific storage of $1.022 \times 10^{-6} \text{ ft}^{-1}$ and a thickness of 23 ft. How much water would it yield if the head declined an average of 1.75 ft over a circular area with radius of 418 ft? (5 pts)

A confined aquifer has a specific storage of $4.033 \times 10^{-4} \text{ ft}^{-1}$ and a porosity of 0.274. The compressibility of water is $4.6 \times 10^{-10} \text{ m}^2/\text{N}$. What is the compressibility of the aquifer skeleton? (5 pts)

1.3 Darcy's Law

An unconfined aquifer has a hydraulic conductivity of 8.7×10^{-2} cm/s. There are two observation wells 597 ft apart. Both penetrate the aquifer to the bottom. In one observation well, the water stands 28.9 ft above the bottom, and in the other it is 26.2 ft above the bottom. What is the discharge per 100-ft wide strip of the aquifer in cubic ft per day? (10 pts)

What is the water table elevation at a point midway between the two observation wells? (10 pts)

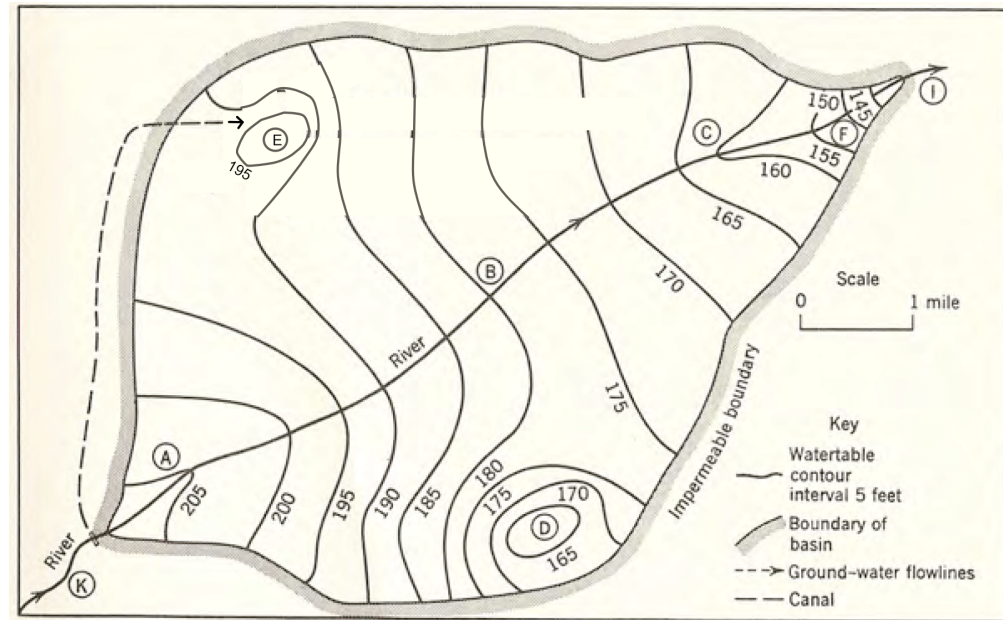
1.4 Groundwater Flow

Write out the full 3-D, transient, heterogeneous, and anisotropic groundwater flow equation. Using plain language, label what each part means. (10 pts)

Write this equation for 2-D, steady-state, isotropic, heterogeneous flow. (10 pts)

1.5 Regional Groundwater Flow

On the below water table plot, draw dashed flowlines indicating local and regional groundwater flow direction. (12 pts)



Name what process, feature, or water balance component each letter is indicating. (8 pts)

- A:
- B:
- C:
- D:
- E:
- F:
- K:
- I: